LAB SAFETY

SCENARIO ANALYSIS

MAKE THE SAFE CALL

WHAT STUDENTS DO

Analyze 10 original laboratory situations, identify evidence of risk, choose the safest first action, and justify decisions using a consistent safety framework.

GRADES 9–11	CORE 45–55 MIN	PRINT + EDITABLE
CORE + HONORS	EXIT TICKET	FULL KEY

Original content • Printer-friendly • Structured reasoning

Welcome, Teacher

Use this ready-to-teach lesson for direct instruction, collaborative practice, a sub plan, or a readiness check.

LEARNING TARGET

I can notice evidence, describe the risk, choose the safest immediate action, and explain why it reduces harm.

SUGGESTED 50-MINUTE FLOW

Time	Action
5 min	Launch: What makes an action the safest first move?
8 min	Model Scenario 0 using Notice Risk Act Explain.
20 min	Pairs complete Scenarios 1-6; teacher checks reasoning.
12 min	Independent Scenarios 7-10.
5 min	Exit ticket and debrief. Honors extension adds 15-25 min.

QUICK START / SUB PLAN

Assign the Student Tool, all scenario pages, and the exit ticket. Students read the four-step framework independently and complete in order. Review with the key next class; a substitute should not improvise site-specific safety procedures.

DIFFICULTY PATH

Support 1/3: Scenarios 1-6 + category bank (35-45 min) | Core 2/3: all 10 (45-55 min) | Honors 3/3: all 10 + extension (60-75 min)

COMMON MISCONCEPTIONS

Reporting only after cleanup; returning excess reagent to stock; treating a colorless liquid as safe; naming a rule without explaining how it reduces the stated risk.

IMPORTANT

This resource supports instruction; it does not replace district policy, teacher supervision, SDS information, or site-specific emergency procedures.

ANSWER KEY

Scenarios 1–4

Reasonable equivalent wording is acceptable when it follows local policy and does not add risk.

1. Goggles on Standby

Evidence: Chemical measuring has begun while Maya's goggles are on her forehead.
Risk: A splash could reach unprotected eyes.
Action: Stop measuring and wear splash goggles correctly before continuing.
Why: The eye barrier must be in place before exposure can occur.

2. Blocked Aisle

Evidence: Backpacks and a cable cross a travel route leading toward the exit.
Risk: Someone could trip, and movement during an emergency could be delayed.
Action: Stop traffic through the area and move the items to the assigned storage location.
Why: Removing the obstruction eliminates the trip hazard and keeps the exit route accessible.

3. Mystery Beaker

Evidence: The beaker has no label, and no one can identify the clear liquid.
Risk: Touching or testing an unknown substance could cause exposure or contamination.
Action: Do not touch or test it; keep others away and notify the teacher immediately.
Why: Avoiding contact prevents additional exposure while the teacher secures and identifies the material by site procedure.

4. Heating Direction

Evidence: The opening of a heated test tube points toward another person.
Risk: Hot liquid or vapor could eject toward that person.
Action: Stop heating; use the teacher-approved holder and point the opening away from all people.
Why: Stopping removes the immediate energy input, and safe orientation reduces direct exposure if material ejects.

ANSWER KEY

Scenarios 5–7

Reasonable equivalent wording is acceptable when it follows local policy and does not add risk.

5. Broken Glass

Evidence: Broken glass is in the sink, and a bare hand is reaching toward it.
Risk: Sharp glass could cut the student.
Action: Do not pick it up by hand; alert the teacher and follow the designated broken-glass procedure.
Why: The designated tools and container prevent direct hand contact and isolate sharp waste.

6. Extra Reagent

Evidence: Previously poured reagent is about to be returned to the stock container.
Risk: The stock chemical could be contaminated, affecting later safety and results.
Action: Do not return it; ask the teacher for the correct waste procedure.
Why: Keeping used material out of the stock bottle protects the shared supply from contamination.

7. Hair Near Heat

Evidence: Loose hair and clothing are near a burner setup.
Risk: Hair or clothing could contact heat or flame and ignite or cause a burn.
Action: Turn off or do not light the heat source; secure hair and loose clothing first.
Why: Removing the ignition exposure before heating prevents contact with flame or hot equipment.

ANSWER KEY

Scenarios 8–10 + extension

Reasonable equivalent wording is acceptable when it follows local policy and does not add risk.

8. Direct Smell

Evidence: The open container is being raised directly toward the student's nose without instruction.
Risk: A concentrated vapor could be inhaled.
Action: Move away, close the container if safe and directed, and ask the teacher; only waft when specifically instructed.
Why: Increasing distance and following teacher direction limits unnecessary inhalation exposure.

9. Splash Report

Evidence: A chemical reached skin, but the student only wiped it and did not report the exposure.
Risk: Chemical may remain on the skin, and delaying the correct response could increase injury.
Action: Alert the teacher immediately while beginning the room's designated skin-exposure rinse, unless the posted site/SDS procedure specifies otherwise.
Why: Rapid rinsing limits contact time while immediate reporting activates the correct site-specific response.

10. Hot Plate Left On

Evidence: An energized hot plate is unattended beside paper.
Risk: Someone could be burned, or nearby combustible material could ignite.
Action: Remain at the station. If trained and it is safe, use the normal control to turn off the hot plate; keep others clear and alert the teacher if it cannot be shut down safely.
Why: Immediate safe shutdown removes the heat source, while guarding the area prevents contact or ignition during escalation.

CORE RUBRIC - 3 POINTS PER SCENARIO

1 point: evidence and risk are specific and connected. 1 point: first action is safe, observable, and immediate. 1 point: explanation shows how the action reduces the stated risk.

HONORS EXEMPLAR

One defensible ranking is 4 first, 9 second, and 10 third. Other rankings earn credit when severity, likelihood, and immediacy are supported with scenario evidence.

CER MODEL

Claim: Pausing to check the setup before acting prevents the widest range of incidents. Evidence: Scenario 1 begins chemical handling before goggles are in place, and Scenario 6 returns used reagent toward shared stock. Reasoning: a deliberate check catches both personal-protection and contamination risks before exposure spreads.

EXIT TICKET GUIDANCE

1. Do not touch or test; keep others away and notify the teacher. 2. Specific actions are observable and directly reduce the identified risk. Items 3-4 are formative.

TEACHER REFERENCE

Sources, rights, and use

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ORIGINALITY

All student scenarios, questions, explanations, and layout are original. External sources informed factual accuracy and instructional scope; no source questions or passages were copied or adapted.

• American Chemical Society. Guidelines for Chemical Laboratory Safety in Secondary Schools and RAMP resources. https://www.acs.org/education/policies/middle-and-high-school-chemistry/safety.html

• American Chemical Society. Safety Equipment Guidelines. https://www.acs.org/education/policies/middle-and-high-school-chemistry/classroom-and-lab-facilities/safety-equipment.html

• American Chemical Society. Student Laboratory Code of Conduct. https://institute.acs.org/acs-center/lab-safety/education-training/high-school-labs/student-lab-code-of-conduct.html

• OpenStax Chemistry 2e, Chapter 1, used only for scope and terminology benchmarking. Current English edition license is CC BY-NC-SA 4.0; no content was adapted for this commercial resource. https://openstax.org/books/chemistry-2e/pages/preface

TERMS OF USE

QUALITY CHECK

Before student use, confirm that emergency procedures and disposal language match your school, room, and available safety equipment.